

Proposed Solution Template

Date	30 June 2026
Team ID	XXXX
Project Name	India's agricultural crop production analysis
Maximum Marks	2 Marks

Project Design Phase

S.No.	Parameter	Description
1	Problem Statement	Indian agriculture faces challenges including stagnating yields, climate dependency, and lack of real-time, data-driven decision support for farmers.
2	Idea / Solution	Develop an AI-powered predictive analytics platform that integrates satellite imagery with localized IoT sensor data to forecast crop yields and recommend optimal agricultural schedules.
3	Novelty / Uniqueness	Utilizes a hybrid approach combining high-resolution satellite vegetation indices with ground-level IoT soil moisture data for "hyper-local" accuracy.
4	Social Impact	Empowers farmers with actionable insights, leading to optimized resource usage, reduced costs, and increased income, supporting national food security.
5	Business Model	B2B SaaS model targeting agricultural cooperatives, government agencies, and agri-tech firms via subscription-based predictive dashboards.
6	Scalability	Built on scalable cloud infrastructure, allowing expansion from pilot districts to national-level implementation by integrating additional data streams.

